



Multiple Choice:

1. Indicate the false statement. Transit time in microwave tubes will be reduced if
 - A. the electrodes are brought closer together
 - B. a higher anode current is used
 - C. multiple or coaxial leads are used
 - D. the anode voltage is made larger
2. What allows microwaves to pass in only one direction?
 - A. RF emitter.
 - B. Ferrite isolator.
 - C. Capacitor.
 - D. Varactor-triac.
3. One of the following is unlikely to be used as a pulsed device. It is the
 - A. multicavity
 - B. BWO
 - C. CFA
 - D. TWT
4. Back ward wave oscillator is based on the
 - A. rising sun magnetron
 - B. crossed field amplifier
 - C. coaxial magnetron
 - D. traveling wave tube
5. The cavity magnetron uses strapping to
 - A. prevent mode jumping
 - B. prevent cathode back heating
 - C. ensure bunching
 - D. improve the phase focusing effect
6. The TWT is sometimes preferred to the magnetron as a radar transmitter output tube because it is
 - A. capable of longer duty cycle
 - B. more efficient amplifier
 - C. more broadband
 - D. less noisy
7. A microwave device that allows RF energy to pass through in one direction with very little loss but absorbs RF power in the opposite direction:
 - A. Circulator.
 - B. Wave trap.
 - C. Multiplexer.
 - D. Isolator.
8. A microwave tube amplifier uses an axial magnetic field and a radial electric field. This is the
 - A. reflex klystron
 - B. coaxial magnetron
 - C. traveling wave tube magnetron
 - D. CFA
9. Magnetic field is used in the cavity magnetron to
 - A. prevent anode current in absence of oscillations
 - B. ensure that the oscillations are pulsed
 - C. help in focusing the electron beam, thus preventing spreading
 - D. ensure electrons will orbit around the cathode
10. An isolator:
 - A. Acts as a buffer between a microwave oscillator coupled to a waveguide.
 - B. Acts as a buffer to protect a microwave oscillator from variations in line load changes.
 - C. Shields UHF circuits from RF transfer.
 - D. Acts as a buffer between a microwave oscillator coupled to a waveguide and acts as a buffer to protect a microwave oscillator from variations in line load changes.
11. Indicate the false statement. An advantage of stripline over microstrip is its
 - A. easier integration with semiconductor devices
 - B. lower tendency to radiate
 - C. higher isolation between adjacent circuits
 - D. higher Q
12. Indicate the false statement. An advantage of stripline over waveguides is its
 - A. smaller bulk
 - B. greater bandwidth
 - C. higher power handling capability
 - D. greater compatibility with solid state devices
13. What should you do for safety when operating at UHF and microwave frequencies?
 - A. Make sure the standing wave ratio is low before you conduct a test
 - B. Never use a horizontally polarized antenna
 - C. Keep antenna away from your eyes when RF is applied
 - D. Make sure that an RF leakage filter is installed at the antenna feed point
14. One of the reasons why vacuum tubes eventually fail at microwave frequencies is that their
 - A. noise figure increases
 - B. transit time becomes too short
 - C. shunt capacitive reactance becomes too large
 - D. series inductive reactance becomes too small
15. Magnetron whose oscillating frequency is electronically adjustable over a wide range is called a
 - A. coaxial magnetron
 - B. dither-tuned amplifier
 - C. frequency- agile magnetron
 - D. VTM
16. To avoid difficulties with strapping at high frequencies, the type of cavity structure used in the magnetron is the
 - A. hole and slot
 - B. slot
 - C. vane
 - D. rising sun



17. The multicavity klystron
- A. is not a good low level amplifier because of noise
 - B. has a high repeller voltage to ensure a rapid transit time
 - C. is not suitable for pulsed operation
 - D. needs a long transit time through the buncher cavity to ensure current modulation
18. The primary purpose of the helix in a traveling wave tube is to
- A. prevent the electron beam from spreading in the long tube
 - B. reduce the axial velocity of the RF field
 - C. ensure broadband operation
 - D. reduce noise figure
19. The TWT is sometimes preferred to the multicavity klystron amplifier, because the former
- A. is more efficient
 - B. has a greater bandwidth
 - C. has a higher number of modes
 - D. produces a higher output power
20. Periodic permanent magnet focusing is used with TWTs to
- A. allow pulsed operation
 - B. improve electron bunching
 - C. avoid the bulk of an electromagnet
 - D. allow coupled cavity operation at highest frequencies
21. The transmission system using two ground planes is
- A. microstrip
 - B. elliptical waveguide
 - C. parallel wire line
 - D. stripline
22. A circulator:
- A. Cools DC motors during heavy loads by turning on a cooling fan for the motors' windings.
 - B. Allows two or more antennas to feed one transmitter.
 - C. Allows one antenna to feed two separate microwave transmitters and receivers at the same time.
 - D. Insulates UHF frequencies on transmission lines and helps to reduce skin effect at high frequencies.
23. The transit time in the repeller space of a reflex klystron must be $n + \frac{3}{4}$ cycles to ensure that
- A. electrons are accelerated by the gap voltages on their return
 - B. returning electron give energy to gap oscillations
 - C. it is equal to the period of the cavity oscillations
 - D. the repeller is not damaged by the striking electrons
24. The attenuator is used in the traveling wave tube to
- A. help bunching
 - B. prevent oscillations
 - C. prevent saturations
 - D. increase gain
25. Indicate the false statement. Klystron may use intermediate cavities to
- A. prevent the oscillations that occur in two cavity klystron
 - B. increase the bandwidth of the device
 - C. improve the power gain
 - D. increase the efficiency of the klystron
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